SONGHE WANG

he/him/his | sxw5765@psu.edu | 814-251-2660 | W362, Westgate Building, University Park, PA, 16802

EDUCATION

The Pennsylvania State University — State College, PA
Ph.D. Student in Computer Science — Aug. 2021 - Current
No.1 in 2023 Spring PhD Qualifying Exam (System, Architecture, Algorithm)

University of North Carolina at Chapel Hill — Chapel Hill, NC
B.S. in Computer Science & Mathematics — Aug. 2017 - May 2021
Deans List

RESEARCH INTERESTS

Computer Vision, Adversarial Machine Learning, Large Language Model, Multi-Agents

SELECTED PUBLICATION

- **S. Wang**, David Jonathan Miller. *When 3D Gaussian Splatting Recovers Real Surfaces.* European Conference on Computer Vision (**ECCV**). 2026
- **S. Wang**, David Jonathan Miller. *Deformable Contact-Aware 3D Object Placement.* Neural Information Processing Systems. (**NeurIPS**). 2026
- **S. Wang**, X. Li, R. Huang, M. Gowda, G. Kesidis. *Temporal-Distributed Backdoor Attack Against Video Based Action Recognition.* Association for the Advancement of Artificial Intelligence (**AAAI**). 2024
- **S. Wang**, Zheng Bao, Jintong E. *Armor: A Benchmark for Meta-evaluation of Artificial Music.* ACM International Conference on Multimedia (**ACMMM**). 2021
- **S. Wang**, K Wei, L Lin, W Li. *A Spatial-Temporal Analysis of COVID-19's Impact on Human Mobility: The Case of the United States.* International Conference of Transportation Professionals (**CICTP**). 2020
- X. Li, **S. Wang**, C. Wu, H. Zhou, J. Wang. *Backdoor Threats from Compromised Foundation Models to Federated Learning.* International Workshop on Federated Learning in the Age of Foundation Models in Conjunction with (**NeurIPS**). 2023.
- Y Nie, **S. Wang**, M Bansal *Revealing the importance of semantic retrieval for machine reading at scale.* Empirical Methods in Natural Language Processing (**EMNLP**). 2019

SELECTED RESEARCH PROJECTS

- **Physics-Grounded 3D Vision** — 2024 - Current
  Bridged theory and practice by proving finite-thickness identifiability conditions for 3D Gaussian Splatting and introducing a deformable, contact-aware 3D object placement method; Paper published at ECCV 2026 and NeurIPS 2026.
- **Backdoor Attacks and Defenses** — 2023 - 2024
  Extended backdoor poisoning attack strategies to video action recognition and federated learning under the foundation model + federated learning setting, offering insights into diverse contexts and developing defenses against these attacks. Paper published at AAAI 2024 and FLFM@NeurIPS 2023.
- **Action Recognition** — 2021 - 2023
  Worked on video-based sign language recognition (SLR) and translation (SLT), skeleton-based SLR, unsupervised sign words clustering, large language model (LLM) assisted SLT, and the security of SLR.
- **Semantic Retrieval** — 2018 - 2019
  Introduced an optimized pipeline emphasizing hierarchical semantic retrieval, achieving state-of-the-art results on the leaderboard test sets of both FEVER and HOTPOTQA. Paper published at EMNLP 2019.

- **Music Generation Evaluation** 2020 - 2021
  Proposed a meta-evaluation benchmark for AI-generated music and highlighted discrepancies between current evaluation metrics and human judgment with extensive experiments. Paper published at ACMMM.
- **Intelligent Transportation** 2020 - 2021
  Analyzed U.S. transportation dynamics during the COVID-19 pandemic using graph theory and detected distinct spatial and temporal patterns in transportation. Paper published at CICTP 2020.

INTERN EXPERIENCE

Research Scientist, BoostDraft, Remote Sept 2024 - Current

- Created an LLM-powered multi-agent orchestration system that extracts structured data from legal clauses, infers schemas, and dynamically answers natural-language queries via synthesized SQL.
- Built legal document alignment, clustering and term retrieval pipelines that all achieve SOTA results on CUAD and Legal NLI.

Research Scientist Engineer, Intuit, CA May 2024 - August 2024

- Research scientist engineer specializing in the training and deployment of Large Language Models (LLMs) and Vision Language Models (VLMs).
- Finetuned Llama2-7b, Mistral-7b and Qwen-VL and got SOTA structured outputs on internal financial documents benchmarks

Research Assistant, Yale University May 2020 - December 2020

- Investigated the task of adapting the existing Text-to-SQL semantic parsers to unknown domains
- Identified several challenges and systematically diagnose these challenges and find that this cross-domain adaptation problem is challenging and far from being solved. Paper submitted to NAACL 2021.

ACHIEVEMENTS

- Chinese National Second-Level Athlete, Professional Go Player
- Carolina Data Challenge 1nd place
- Putnam Mathematics Competition top 1% nationwide
- Virginia Tech Math Competition top 5% nationwide

PROFESSIONAL SERVICE

- Reviewer, Association for the Advancement of Artificial Intelligence (AAAI), 2022 - 2023
- Reviewer, Empirical Methods in Natural Language Processing (EMNLP). 2020
- Reviewer, Proceedings of the First Workshop on Interactive and Executable Semantic Parsing

TEACHING EXPERIENCE

- **Teaching Assistant** 2023 Spring
  CSE 597: Vision and Language
  Instructor: Prof. Huijuan Xu
- **Teaching Assistant** 2022 Fall
  CMPSC497: Deep Learning for Computer Vision
  Instructor: Prof. Huijuan Xu

PROGRAMMING SKILLS

Programming Languages: Python, C, C++, C#, MATLAB, Java, JavaScript, HTML, Swift, SQL, CSS
Tools: PyTorch, TensorFlow, OpenCV, Colab, CUDA, Docker, MySQL, SciPy, Hadoop, Git, HuggingFace